Nilabja Ghosh

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Work Experience

Ernst & Young LLP (EY India) **May 2025 – Present**
Consultant- Data Scientist *Kolkata, West Bengal*

- Developed and managed demand forecasting models for consumer goods for the Europe region using **Lasso**, **XGBoost**, and **Random Forest** in PySpark on Azure Databricks, improving forecast accuracy and **reducing stockouts**.
- Engineered features to capture month-end sales spikes in alignment with demand planner requirements, improving KPIs - **forecast accuracy** by **+5%** and **forecast bias** by **-2%** across all business categories.
- Processed and analyzed large-scale datasets (**historical sales**, **market trends**, **seasonal patterns**) and conducted root cause analysis (RCA) using **SQL**/**Pyspark** to identify issues and enhance machine learning model's performance.

Feedsense AI Pvt. Ltd. (Previously- Vista Intelligence Pvt. Ltd.) **Jul 2023 – April 2025**
Quantitative Research Executive *Kolkata, West Bengal*

- Research on **Black Scholes** and **Heston** model for extraction and prediction of implied volatility to used it in designing a prediction-based options trading strategy for the **BankNifty** index.
- Developed and implemented an investment and hedged strategy for Axis Bank's market price, delivering an average annual return of **43%**, consistently outperforming the market by **93%** over the past 5 years, while minimizing risk exposure and reducing drawdown by **14%**.
- Engineered an efficient real-time video transcription model using OpenAI's Whisper, seamlessly integrated with Ollama-3.2 for immediate sentiment analysis, enabling rapid news content understanding with a directional accuracy of **65%**.

Research & Publication

- **StereoHate**: Toward identifying stereotypical bias and target group in hate speech detection. *DOI: 10.1017/nlp.2024.29*

Key Projects

A Model for Explaining Toxic Text in Indian Languages. **May, 2022 – Dec, 2022**
Summer project at Department of Computer Science and Engineering, ***IIT Patna*** *Patna*

- Research on different types knowledge graphs and bases to understand the toxic text and generate the corresponding stereotypes.
- Multilingual Bidirectional and Auto-Regressive Transformer (**mBART**) is used to generate the stereotype and target group from the given Hindi post.

Image Classification using Self-Supervised Learning. **May, 2022 – Jul, 2022**

- 10% data is sampled from the CIFAR-10 dataset using balanced sampling. This data is used for extracting features using Barlow Twins method and **Res-Net 50** architecture and Logistic Regression is used to classify the images.
- **65% accuracy** is achieved against 15% accuracy in supervised learning with 10% samples.

Detection of Malaria from blood smears using CNN: A comparative study of optimizers. **May, 2022 – Jul, 2022**

- We use a 3-layer **CNN** model to classify blood smears into infected and non-infected ones, using Stochastic Gradient Descent with momentum, Adam, Adagrad, RMSProp optimizers.
- The **SGD with momentum** optimizer achieved a test accuracy of **96.11%** with a validation loss of **0.4061**. It was closely followed by **Adam**, which reached a test accuracy of **95.84%**, and **Adagrad**, with a test accuracy of **95.62%**.

Certifications

- Securities and Exchange Board of India **(SEBI)** Certified NISM Series XV **Research Analyst**.

Technical Skills & Interests

Languages: Python, MySQL/ PostgreSQL, R Programming Language
Libraries: NumPy, Pandas, PyTorch, Matplotlib, PySpark, LangChain
R packages: ggplot2, dplyr, tidyvers. **Tools:** MS Powerpoint, MS WORD, MS Excel, LaTeX
Area of Interest: Machine Learning, Natural Language Processing, Deep Learning, Statistics, Demand Forecasting, Quantitative Analysis

Education

Year	Degree	Institution	Performance (CPI/%)
2023	M.Sc. in Big Data Analytics	RKMVERI	CGPA: 7.18/10
2021	B.Sc. in Physics Hons.	St. Xavier's College, Kolkata	CGPA: 7.5/10
2018	Class XII- ISC	Holy Child School	% Marks: 91.75
2016	Class X - ICSE	Holy Child School	% Marks: 92.8

Awards and Achievements

- Received *"Client Extraordinary"* Award at EY for delivering exceptional client value and serving as a trusted business advisor.
- Runners-up at *"All-in Champions League"* football tournament organised by EY India.